
Monte Carlo Counterparty Credit Risk Engine

Final Client Readout -- ESF Derivatives Book

Berkeley MFE Industry Project | Prepared for Capitolis Risk, Quant & Technology

Valuation date 2026-08-28 | September 2026

Portfolio MPE99 of \$51.0M on the Brief's Close-Out Definition

\$51.0M

Portfolio MPE99 (close-out)

peak 2026-09-20, 99th pct

\$11.6M

Peak Expected Exposure

close-out, within 1 year

\$129.8M

Current Exposure (t=0)

uncollateralized, no margin

\$11.7k

CVA (close-out, BBB proxy)

SA-CVA capital \$0.10M

- What we built: a validated Monte Carlo engine simulating USD short rate (Hull-White 1F), JPY short rate, 37 equities and USDJPY (correlated GBM), repricing all 16 trades at every scenario and date
- Headline result: portfolio MPE99 \$51.0M on the brief's definition -- 78% of the sum of the three counterparties' own MPE99s, because netting never crosses counterparties and the books peak within days of each other
- Validated: t=0 self-check to 0.0000%, martingale tests, delta-normal VaR benchmark (1.04x), Kupiec backtest, 142 automated tests
- Key conclusion: risk is short-dated (~4 months) and concentrated in one \$500M bond forward (BF_0003) at CPTY_C -- a limit or collateral on that single trade would move the portfolio number more than any modelling choice in this report

INTRO

The Book: 16 Trades, Dominated in Size by One \$500M Bond Forward

Trade	Type	Cpty	Ends	NPV today
EQTRS_0001	Equity TRS	A	2026-10-14	-\$5,547,056
EQTRS_0002	Equity TRS	A	2026-10-09	-\$1,076,056
EQTRS_0003	Equity TRS	A	2026-10-16	\$9,921,581
EQTRS_0004	Equity TRS	B	2026-11-02	-\$873,451
EQTRS_0005	Equity TRS	B	2026-10-09	\$148,605
EQTRS_0006	Equity TRS	B	2026-10-30	\$1,629,065
EQTRS_0007	Equity TRS	C	2026-10-23	-\$373,630
EQTRS_0008	Equity TRS	C	2027-07-22	\$2,217,998
BF_0001	Bond Forward	A	2026-10-30	\$512,531
BF_0002	Bond Forward	A	2026-11-08	-\$357,505
BF_0003	Bond Forward	C	2026-12-06	\$121,262,854
BF_0004	Bond Forward	C	2026-10-25	\$1,973,388
BTRS_0001	Bond TRS	A	2028-01-15	\$119,460
BTRS_0002	Bond TRS	B	2027-01-15	\$210,278
BTRS_0003	Bond TRS	B	2027-04-09	\$28,844
BTRS_0004	Bond TRS	C	2026-11-04	\$36,376



Life span of every trade: nearly all exposure disappears within 4 months.

WHY ONE TRADE DOMINATES

- BF_0003: 500M notional short forward on a 2.88% 2049 Treasury, struck at 100, trading ~25pts below par

- This single position is why CPTY_C drives concentration, SA-CCR EAD and CVA throughout the results

INTRO

16 Trades, 3 Counterparties, 37 Equities and USDJPY, One Engine

Instrument	Count	Counterparties	Risk factors
Equity TRS	8 (incl. 1 JPY compo)	A, B, C	USD curve + equity spots (+ USDJPY for compo)
Bond Forward	4	A, C	USD curve only (risk-free bonds)
Bond TRS	4	A, B, C	USD curve only

- 37 unique equities (31 USD, 6 JPY) + USDJPY + 5 Treasury bonds
- Capitolis is strictly the seller: falling market values increase counterparty exposure to us

Today's agenda

- The brief and what we delivered against it • Data sourcing • Pipeline and architecture
- Modelling choices • Trade-offs and alternatives considered • Feedback incorporated
- Results (the core of this deck) • Conclusions and next steps



Trade life spans: most of the book matures within ~4 months.



What Capitolis Gave Us, and Asked For

The brief, the deliverables, and how every ask maps to what we built

The Brief: No Initial Margin, 10-Day Close-Out, Capitolis Is Seller

Deliverables (kickoff, slide 16)

- Simulation engine codebase: modular Python, extensible to new asset classes / model types
- Technical report: calibration methodology, model choices, key results, production-hardening suggestions, performance/convergence analysis

Key assumptions specified by Capitolis (slide 9)

- No initial margin
- Variation margin on date t = NPV on the prior day on the path
- 10-business-day close-out period
- Scenario count is our choice; must evidence convergence
- Falling market values increase counterparty exposure to us (we are strictly the seller)

4 core objectives (kickoff, slide 5)

Objective	Requirement
1. Stochastic models	Risk factor models (our choice), simulate correlated paths; defend every decision
2. Pricing integration	Feed simulated market states into the supplied pricer library to generate scenario NPVs
3. Exposure and Greeks	PFE, MPE, EE profiles + efficient Greeks across scenarios. Extra credit: xVA
4. Performance	Optimise the engine for speed and scalability

Every Ask Mapped to What We Delivered, Including Both Extra-Credit Items

Ask	Delivered	Evidence
Stochastic models for rates, equity, FX	Hull-White 1F (+G2++ alternative); correlated GBM + quanto; JPY factor	Sections 4, 6
Pricing model integration	Every scenario repriced via the supplied capitolis_pricers library, unmodified	Section 2.2
Exposure profiles: PFE, MPE, EE	Close-out (headline) + uncollateralized level, by trade/counterparty/portfolio	Section 7
Efficient Greeks	CRN bump-and-reprice + pathwise cross-check + exact analytic rate Greek	Section 9
Performance / scalability	10.6x curve speed-up, 7.7x combined; multiprocessing across scenarios	Section 5.3
Extra credit: xVA	CVA, DVA, FVA, KVA; SA-CVA capital; SA-CCR EAD -- on both exposure conventions	Section 8
Extra credit: risky bonds + CDS	Risky-bond sample trade on a bond-index issuer-curve proxy (no CDS data obtainable)	Section 8.9



Data Sourcing

Every input, what it is used for, how it was cleaned, and what we assumed

Real Market Data Everywhere Verifiable; Proxies Documented Where Not

Input	Source	Used for	Status
USD SOFR futures (33 live)	Databento (CME)	Discount curve, HW1F fit	Live, cross-checked vs Treasury.gov
37 equity spots & dividends, USDJPY	yfinance	GBM start values & drift, vol, correlation	Live, keyed by ISIN
39 realised volatilities	3y daily returns (above)	Diffusion size for GBM/HW1F	Proxy: no options data available
39x39 correlation matrix	616 aligned daily dates	Cholesky / PCA simulation	Static; USD row on 10y-yield changes
USD curve long end, swaption cubes	Bloomberg one-time export (2026-08-31)	Curve splice beyond 6.5y; mean reversion	Licensed: derived numbers only
JPY OIS history (35 tenors, 2011-26)	Bloomberg, provided by project team	JPY curve, JPY factor vol, mean-rev. test	Licensed: derived numbers only
Treasury CMT yields, TONA, JGB yields	FRED / Bank of Japan / MOF	Rate vol, backtest, G2++ calibration	Public

Two Real Bugs Found by Cross-Checking Against Independent Sources

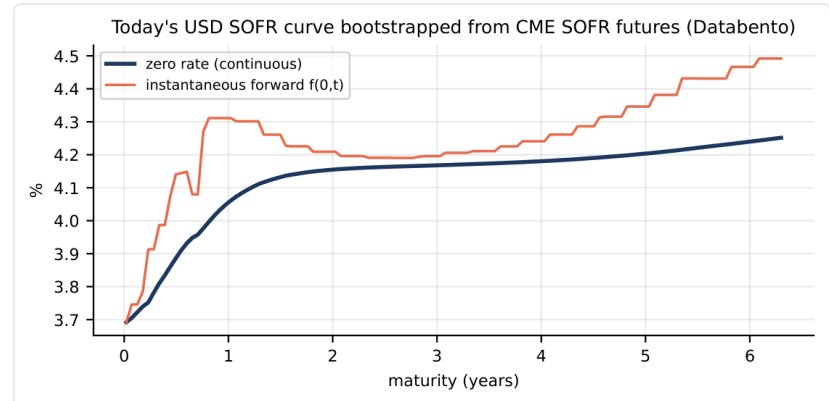
Issue found	How caught	Fix / impact
USD curve extrapolated flat beyond ~6.5y futures range	Repricing BF_0003 on Bloomberg's own zero curve	Bloomberg long-end splice; BF_0003 NPV moved \$101.4M -> \$121.3M (within 0.4% of Bloomberg reprice)
USDJPY drift had the wrong sign	Deriving JPY drifts from the martingale condition	JPY names were drifting ~2x2.6%/yr too low in USD; corrected, quanto term added
USD rate sigma from overnight SOFR understated 2-30y yield moves	10-day realised vs modelled yield-move comparison	sigma refit to realised 2y-30y Treasury yield vol: 63bp -> 96bp
SOFR futures bootstrap: serial-future wraparound gap	Cross-check vs Treasury.gov par curve (73bp vs 56bp at 10Y)	400-day plausibility threshold; curve coverage 3.6y -> 6.3y

- Data quality checks throughout: self-check $EE(0)$ vs direct exposure to 0.0000%; Hull-White curve refit to $1e-9$; positive semi-definite correlation matrix verified
- Assumptions disclosed, not hidden: counterparty rating BBB, LGD 60%, static correlation, JPY mean reversion at its lower bound (every real calibration route gave a negative value)

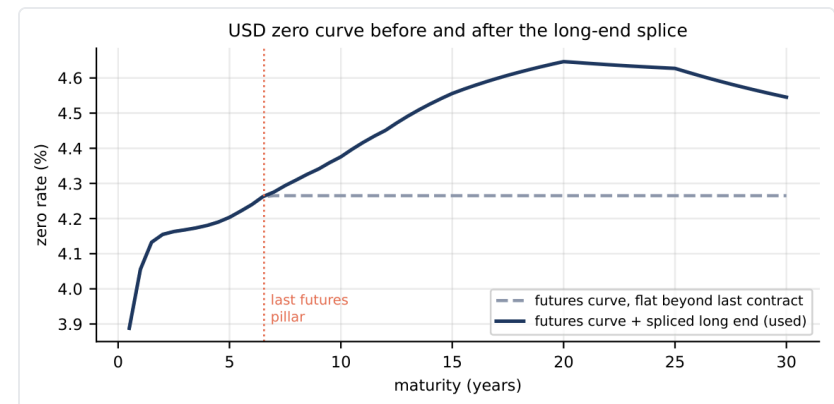
The USD Curve: 33 Live Futures Bootstrapped, Spliced at the Long End

WHY SPLICING, NOT EXTRAPOLATING FLAT

- A 22-year bond has duration ~ 13 -- a flat long end understates discounting of exactly the trade (BF_0003) that dominates the book
- The splice is joined continuously at the last futures pillar: nothing inside the live futures range changes, only the part we had no market data for
- Validated independently: BF_0003's NPV on the spliced curve (\$121.3M) lands within 0.4% of repricing it directly on Bloomberg's own zero curve
- Licensed Bloomberg data is used only to extend the shape beyond the live futures range -- we never redistribute the raw licensed numbers, only this derived splice



Bootstrapped USD SOFR curve: zero rate and instantaneous forward, 0-6.5y.

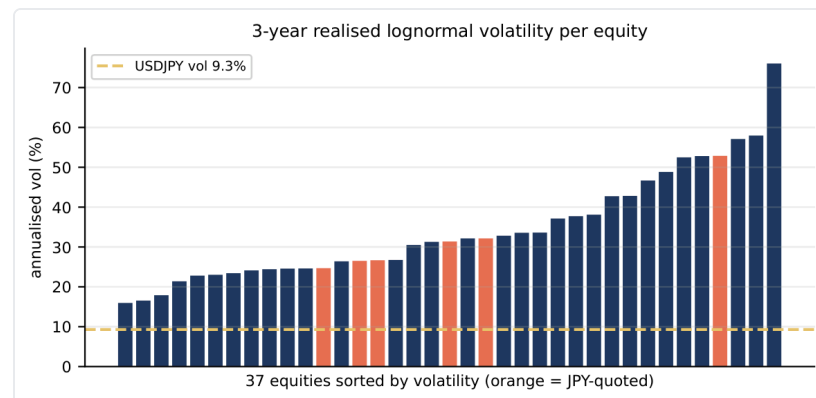


Zero curve before (flat) and after the Bloomberg long-end splice. Identical inside the futures range.

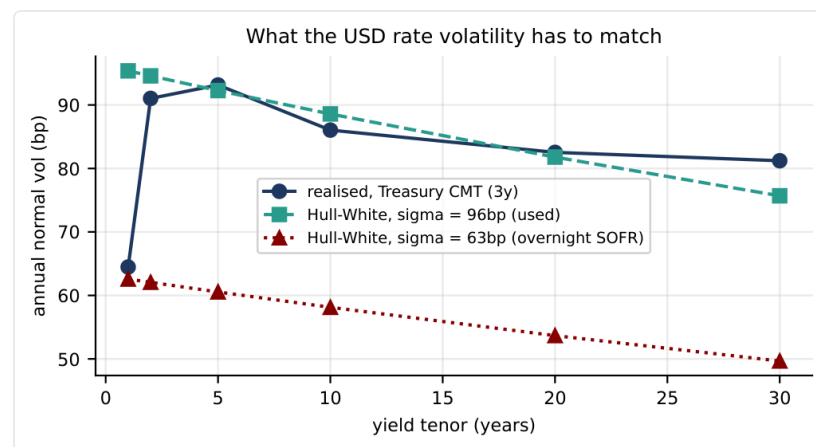
Volatility: 3-Year Realised, Fitted to Where the Book's Risk Actually Lives

WHY NO OPTIONS DATA AVAILABLE

- Implied vol is the pricing-industry norm, but we had no single-name options data (Bloomberg export has only SPX/TOPIX index vols, which would need a per-name basis assumption)
- 3-year realised is a reproducible, documented proxy; its backward-looking limitation is disclosed and stress-tested directly (vols $\times 1.25/\times 2$ in Section 7.9)
- 3 years balances responsiveness against noise: long enough to average out idiosyncratic single-day jumps, short enough to reflect the current vol regime rather than a decade-old one
- The same convention is applied to every one of the 37 names, so the cross-section stays comparable rather than mixing proxy quality name by name



3-year realised lognormal volatility per equity (orange = JPY-quoted).

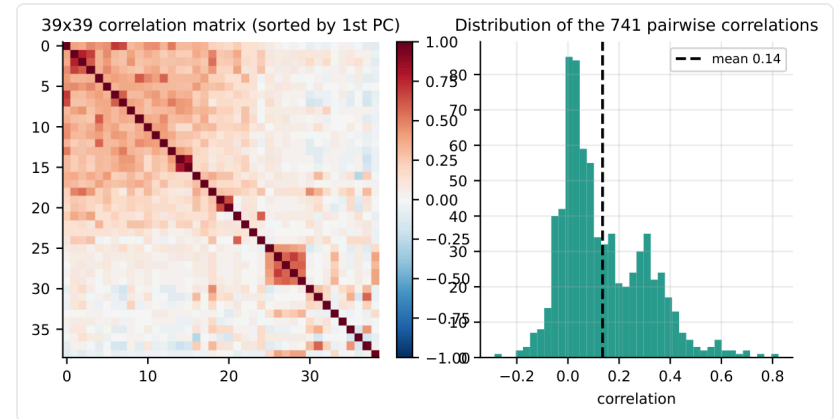


Realised Treasury yield vol by tenor vs. the Hull-White implied shape for both sigma choices.

DATA

Correlation: One Static 39x39 Matrix From 616 Aligned Trading Days

- Assumed constant over the one-year simulation horizon -- correlations tend to rise in stress; this is stressed directly in the model-risk study (Section 7.9: +30% toward 1 moves portfolio MPE99 by +22%)
- A tiny eigenvalue clip guards against CSV round-off making the matrix numerically non-positive-semi-definite before Cholesky factorisation



Left: 39x39 matrix reordered by first principal component. Right: distribution of the 741 pairwise correlations.

Every Assumption Not Sourced From Data, Stated Explicitly

Assumption	Value used	Where it matters	Sensitivity shown
Counterparty credit rating	BBB (proxy, all 3 counterparties)	CVA, DVA, FVA, SA-CVA capital	AAA \$5.0k to B \$32.8k (2.6x AA-BB range)
Loss given default (LGD)	60% (40% recovery)	CVA, DVA default-probability triangle	Market-consensus value for senior unsecured
Margin period of risk (MPOR)	10 business days	Close-out exposure window	Per the brief; not varied
Collateral / CSA terms	None (uncollateralized)	Headline vs. level exposure choice	MPOR illustration: CPTY_C moves 82% if margined
Correlation structure	Static over 1-year horizon	All joint simulation	+30% toward 1: portfolio MPE99 +22%
JPY mean reversion	0.001 (lower bound)	JPY factor dynamics	Every real calibration route gave a negative value
Own credit / funding spread	= BBB counterparty proxy curve	DVA, FVA	Shown across AA to BB in Section 8.5



Pipeline and Architecture

Six stages, data to outputs; modular, reproducible, and fast

PIPELINE

Six Stages From Real Market Data to Every Risk Output



Speed: three bottlenecks fixed

Bottleneck	Fix	Before	After
Discount curve per (scenario, node)	Analytic Hull-White bond price, not a Curve object	0.206 ms/call	0.019 ms/call (10.6x, exact)
Pure-Python pricers (~1.2ms each)	Multiprocessing over scenarios, 8 workers	113 ms/scenario	51 ms/scenario
Combined, 2,000 scenarios x 16 trades	Both fixes together	~12 min projected	93s (7.7x)

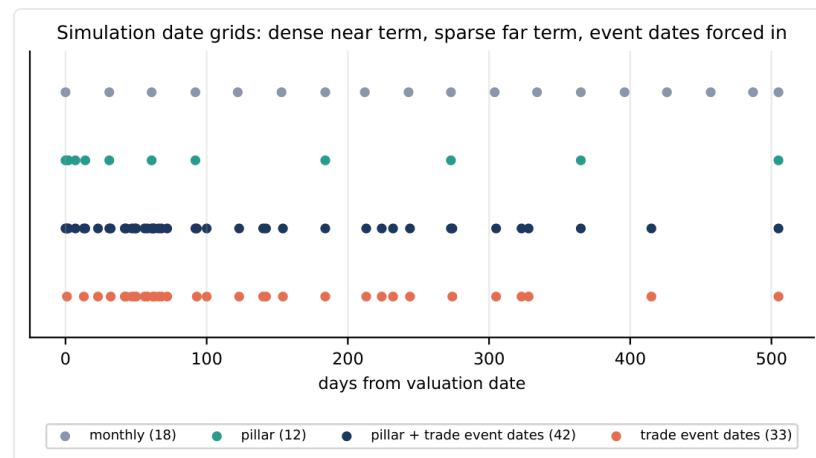
PIPELINE

Simulation Dates: Pillar Tenors Plus Every Trade's Own Event Date

Grid	Node count	What it does
Monthly	18	Simple, but spends nodes evenly and rarely lands on a trade's own dates
Pillar dates only	12	Standard market tenors (O/N..10Y); dense short-term, sparse long-term
Pillar + trade event dates (used)	42	Pillar tenors plus every trade's reset/settlement/maturity forced in
Trade event dates only	33	Exact on cash flows, but no systematic curve-tenor coverage

WHY PILLAR + TRADE EVENT DATES

- Industry convention (pillar tenors) plus exactness on this book's own cash flows -- neither alone is enough
- Dense near-term, where the close-out window and most trade maturities sit, matches where risk actually changes fastest
- A trade settling inside a close-out window is excluded from both legs of that window -- forcing its date onto the grid is what makes that exclusion possible



Four candidate grids for this book: dense near-term, sparse far-term, with event dates forced in.

Building the Engine, Steps 1-4: From Raw Data to Calibrated Models

1

Repository & data layer

Ingest Capitolis-supplied trades/pricers; pull SOFR futures, equity/FX spots, Treasury/JPY/BOJ/MOF history

```
market/
sofr, equities, fx, vols,
correlations, boj, mof_jgb,
bloomberg, credit_spreads
```

2

Market data assembly

Bootstrap the USD curve, compute realised vols, build the 39x39 correlation matrix, proxy credit spreads

```
market/sofr.py
market/equities.py
market/credit_spreads.py
```

3

Risk-factor models

Implement Hull-White 1F (+G2++), correlated GBM for equity/FX, the JPY Hull-White factor, PCA factor model

```
models/
rates, g2pp, equity_fx,
equity_factor_model, credit
```

4

Calibration

Fit sigma/a to swaption cube and realised yield vol; fit JPY factor to the real OIS history; least-squares G2++

```
models/
calibration, hw_calibration
```

5

Simulation engine

Build the date grid (pillars + trade events + MPOR nodes), draw correlated shocks, step every factor jointly

```
simulation/
engine, random_numbers,
parallel
```

Design principle -- modularity: each package has one job and is independently testable (the 142 tests target these modules directly).

Building the Engine, Steps 5-8: Pricing Through to Validation

6

Pricing integration

At every (scenario, node): build a MarketState from simulated rates/spots/FX, call the unmodified pricer library

capitolis_pricers
(given, independently
reviewed, never modified)

7

Aggregation & exposure

Net by counterparty, apply the close-out/level exposure definitions, compute EE/median PFE/PFE99/MPE

exposure/
spec_exposure, aggregate,
collateral

8

Greeks

Bump-and-reprice with common random numbers; pathwise cross-check for equity/FX; exact analytic rate Greek

greeks/
book, exposure, pathwise,
bumps

9

Credit risk & capital

CVA/DVA/FVA/KVA on the same paths; Basel SA-CVA capital; SA-CCR EAD; the risky-bond extra-credit trade

exposure/
cva, xva, sa_cva, sa_ccr

10

Stress, backtest & tests

13 stress scenarios; Kupiec/Christoffersen backtests against realised history; 142 automated unit/statistical tests

stress/scenarios.py
validation/backtest.py
tests/ (142 tests)

Design principle -- reproducibility: every number regenerates from one command per stage (Appendix A10), nothing is one-off or hand-tuned.

IE

Modelling Choices

What we implemented, with the exact parameters used

MODELS

USD Short Rate: Hull-White 1F, Fitted Exactly to Today's Curve

$$dr(t) = [\theta(t) - a r(t)] dt + \sigma dW(t)$$

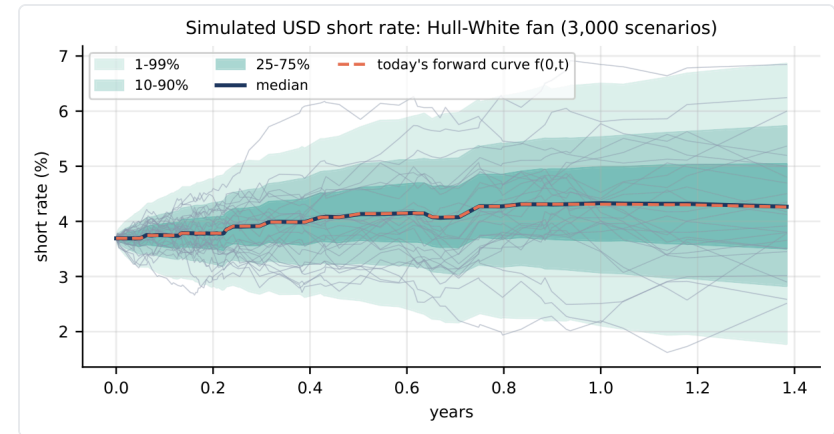
$$r(t) = x(t) + \alpha(t)$$

Parameter	Value	Source
sigma	0.96%	Fitted to realised 2y-30y Treasury yield vol
a (mean reversion)	0.0167	USD swaption cube (1M expiry, 1Y-15Y tenors)
r(0)	3.69%	Bootstrapped SOFR futures curve

WHY HULL-WHITE 1F

- Reproduces today's curve to 1e-9, so discounting is exact at t=0
- Analytic bond price at every node -- 10x faster than a curve object
- Gaussian: supports negative rates, essential for JPY
- Only 2 parameters (a, sigma) to fit and defend -- fewer than the data (a few years of swaption/yield history) can comfortably support
- Industry-standard and well understood, which lowers the model-validation burden for a reviewer
- Rejected: CIR/BK (floored at zero, invalid for JPY); LMM/HJM (too many parameters for our data and slower); constant rate (ignores funding-leg and bond-forward rate risk)

A two-factor alternative (G2++) is implemented, fully calibrated and compared in the next section -- it is not the default because it changes the headline by only -3%.



Simulated USD short rate fan (3,000 scenarios); median tracks today's forward curve.

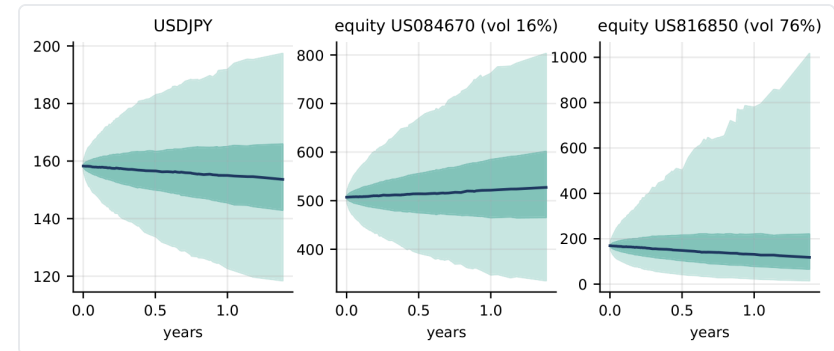
MODELS

Equities, FX and Correlation: GBM Driven by the Simulated Rates

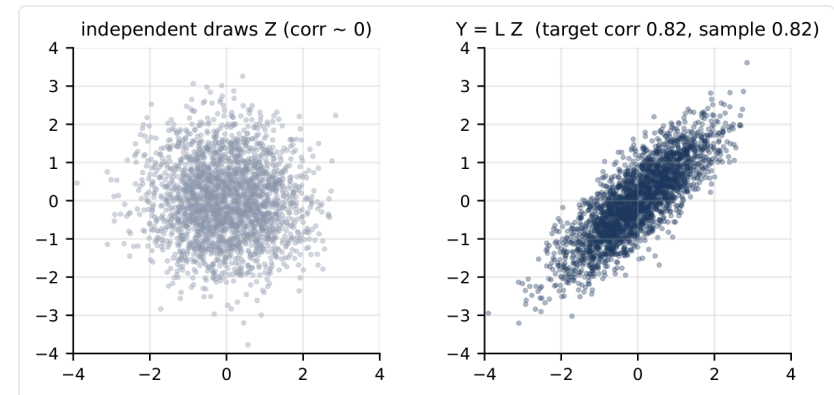
$$dS_i/S_i = (r_{USD} - q_i) dt + \sigma_i dW_i$$
$$dX/X = (r_{JPY} - r_{USD} + \sigma_X^2) dt + \sigma_X dW_X$$

WHY GBM + CHOLESKY

- Matches the lognormal vol convention we measure and the market default for equity CCR; needs only one vol per name
 - Drifts derived from the martingale condition, not assumed -- verified independently by martingale tests
 - No unobservable parameters beyond that single vol -- Heston (vol-of-vol), SABR (beta, rho) and jump models all need inputs we don't have
 - Keeps paths exactly rescalable under a spot bump, which is what later makes equity/FX Greeks ~100x cheaper than naive resimulation (Section 6)
 - Rejected: Heston/SABR (no option surfaces available); jump models (unobservable parameters) -- both disclosed as understating fat tails
-
- 741 pairwise correlations, mean 0.14; first PCA factor (a 'market' mode) explains 20% of variance alone
 - 37 equities span an order of magnitude in vol (a few names at 50-76%), dominating tail exposure of the trades referencing them



Simulated USDJPY and two equities (low-vol vs. highest-vol name), 1-99% and 25-75% bands.



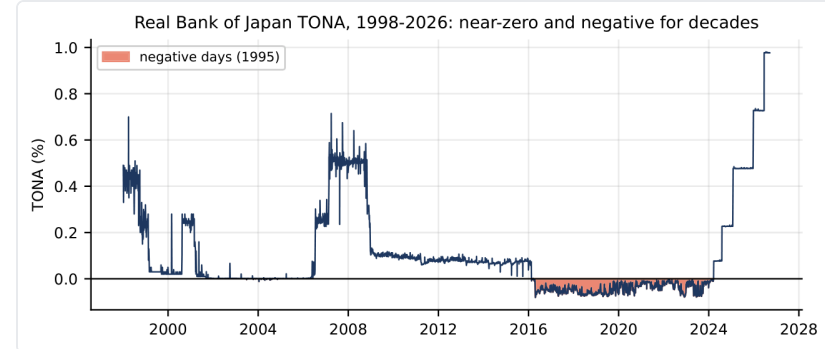
Cholesky demo: independent draws (left) become correlated draws (right), target corr. 0.82.

JPY: a Genuine Negative-Rate-Capable Hull-White Factor

Parameter	Value	Source / status
Curve	JPY OIS zero curve, 2026-08-28	Bootstrapped; matches Bloomberg's own curve to 1bp
sigma	0.267%/yr	Realised vol of the overnight call rate, 3y window
a (mean reversion)	0.0010 (lower bound)	All 3 real JPY calibration routes gave a negative a
USD-JPY rate correlation	-0.040	Real SOFR vs TONA daily changes (n=1983); statistically zero

WHY THE LOWER-BOUND MEAN REVERSION

- For two decades the BOJ pinned the short end (zero / negative policy, YCC), so long tenors moved more freely than short -- the reverse of what a mean-reverting factor assumes
- Confirmed by three independent real data sources (daily OIS history, 52 years of JGB yields, a 2026 swaption cube), every one giving a negative fitted a -- not a one-off artifact of noisy data
- A negative a was rejected outright: it would make the simulated short rate diverge over the horizon, which is both numerically unstable and economically meaningless
- A single Gaussian factor cannot produce the observed shape; $a=0.001$ (the Ho-Lee limit, flat vol across tenors) is the closest a one-factor model can get
- The lower bound keeps the model fully analytic and negative-rate capable, so no other part of the engine needs to change to accommodate it
- Rejected: constant USD-rate differential (cannot represent JPY's own dynamics); a two-factor or regime-dependent model would match the shape but is listed as future work



Real Bank of Japan TONA, 1998-2026: at or below zero for most of 25 years -- a model with a floor could not represent this.

Exposure Built to the Brief's Exact Definition, Not a Simplification

$\text{exposure}(t) = \max(V(t + 10bd) - V(t - 1bd), 0)$, $V(t-1bd)$ = variation margin, signed

- Netting applied to value before the positive part -- a +10M and a -6M trade with the same counterparty give exposure 4M, not 10M
- Netting never crosses counterparties: portfolio exposure is the SUM over counterparties of each one's own $\max(\text{sum of trades}, 0)$
- EE = mean across scenarios; median PFE = 50th pct; PFE99 = 99th pct; MPE = peak of the PFE curve over all dates
- Simulation grid carries 3 nodes around every reporting date ($t-1bd$, t , $t+10bd$) on the same simulated path -- 42 reporting dates for this book
- Standard market pillar dates (O/N to 10Y) used, not a plain monthly grid -- denser near term, where risk changes fastest
- Every trade's own reset/settlement/maturity date forced onto the grid so exposure jumps at cash-flow dates are not smoothed over



Trade-Offs, Alternatives and Enhancements

What we tried and rejected, and why; every choice measured, not asserted

TRADE-OFFS

Latin Hypercube Beats Four Other Sampling Methods on a Controlled Test

	Call RMSE (Test 1)	PFE99 noise (Test 2)	Median noise (Test 2)	Verdict
Pseudo-random	1.0x (baseline)	2.60%	0.66%	Baseline
Antithetic	1.4x better	1.98%	0.39%	Modest, real engine
Moment-matched	6.3x better	2.49%	0.35%	
Sobol	28.7x better	1.64%	0.70%	Best PFE99, no median gain
Latin Hypercube	50.6x better	2.25%	0.42%	Best on both

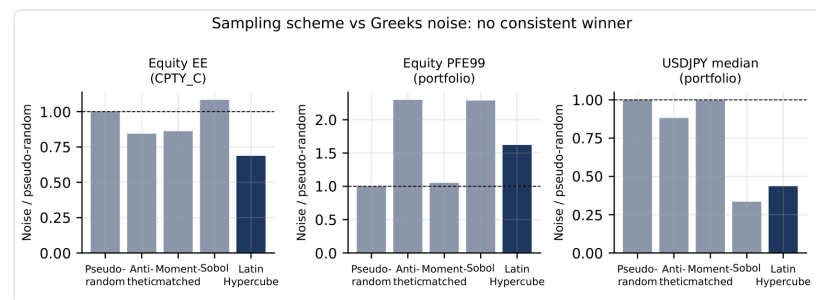
Test 1: European call price vs. Black-Scholes, RMSE relative to pseudo-random. Test 2: relative standard deviation of the real 663-dim engine's estimator across 8 repeats (% of reference value).

✓ SELECTED

WHY LATIN HYPERCUBE

- Best controlled-test error, 50.6x vs pseudo-random
- Only method better than pseudo-random on BOTH real-engine statistics -- the deciding factor
- Negligible extra cost over pseudo-random -- ~40 factors is cheap next to repricing
- Stratifies every marginal so tails are sampled, unlike Sobol's joint-distribution assumptions

- On the real 663-dimensional engine, advantages compress -- no single method wins both PFE99 and the median



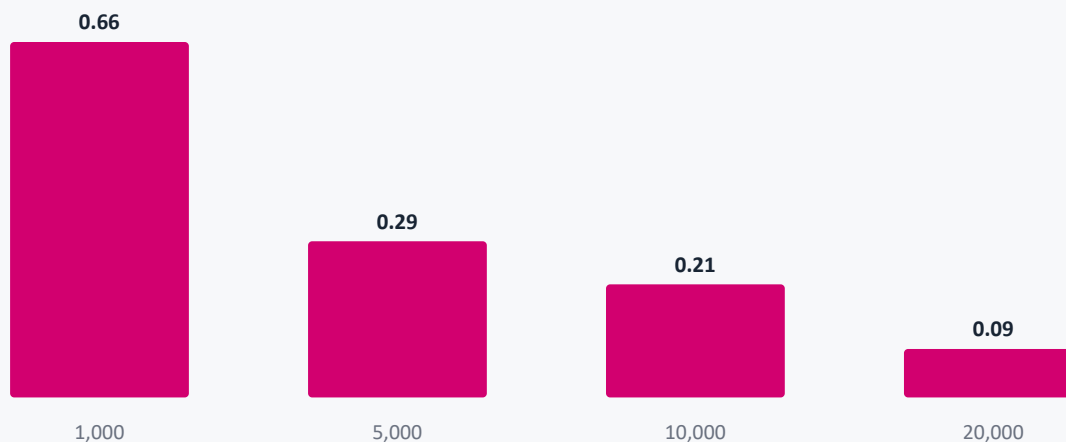
Left: 51x lower RMSE vs Black-Scholes. Right: real-engine estimator noise by method.

- Sobol is the controlled-test runner-up (28.7x) but loses its edge on the median PFE and is less robust at this dimensionality

TRADE-OFFS

Scenario Count: N=5,000 for Reporting, N=10,000 for Limit Sign-Off

Relative standard error of PFE99 falls as $1/\sqrt{N}$



N	Rel. SE	Est. time (8 cores)	Verdict
1,000	0.66%	119s	Iteration / what-if
5,000	0.29%	479s	SELECTED: standard reporting
10,000	0.21%	929s	SELECTED: limit sign-off
20,000+	0.09%	1,829s+	Diminishing returns

WHY 5,000 / 10,000

- Cost grows linearly; error falls only as $1/\sqrt{N}$ -- diminishing returns past ~15-20k
- Remaining sampling error (<0.4%) is an order of magnitude smaller than the model-risk sensitivities (Section 7.9)
- N=5,000 keeps a full reporting run to ~8 minutes on 8 cores -- practical for routine, repeated use, not just a one-off

- Three tiers (1k/5k/10k) match three real use cases -- iteration, standard reporting, limit sign-off -- rather than one setting for everything

TRADE-OFFS

Two-Factor Rates (G2++) Changes the Portfolio MPE99 by Only -3%

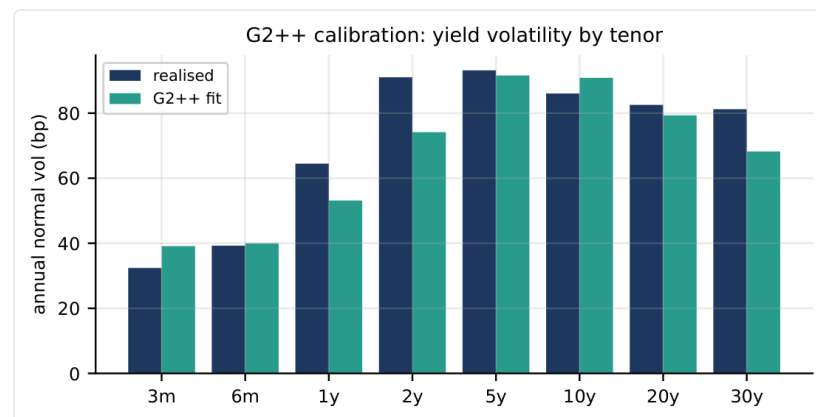
	Curve fit	Negative rates	Calibration error	Params
Hull-White 1F	Exact	Yes	N/A (level only)	2 (a, sigma)

G2++ (2-factor)	Exact	Yes	16% (Frobenius, yield covariance)	5 (a,b,sigma,eta,rho)
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✓ SELECTED

Hull-White 1F (default)

Netting set	MPE99 1F	MPE99 2F	Change
CPTY_A	\$26.2M	\$26.0M	-1%
CPTY_B	\$9.5M	\$9.3M	-3%
CPTY_C	\$28.9M	\$28.9M	-0%
Portfolio	\$51.5M	\$50.0M	-3%



G2++ fit to realised yield-change vol by tenor: matches level/slope, not the 5y hump.

WHY HULL-WHITE 1F OVER G2++

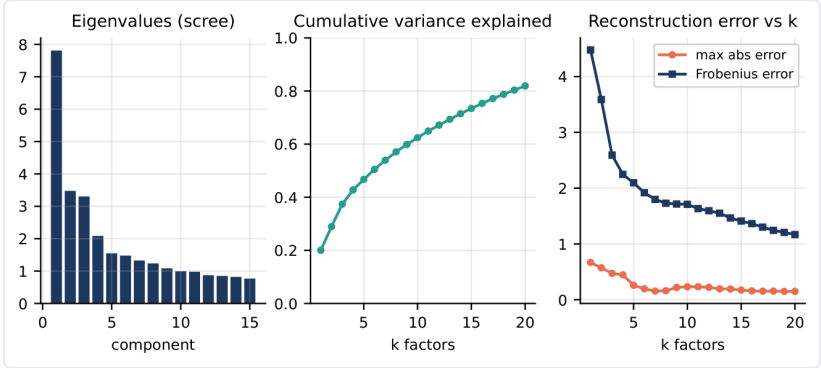
- -3% on the headline doesn't justify 3 extra parameters (2 vs 5) -- no demonstrable accuracy gain for the added complexity
- G2++'s own calibration carries a 16% fit error -- it trades one uncertainty (1F's missing slope factor) for another (2F's imperfect fit), not a clear improvement
- HW1F calibrates to the forward-looking swaption cube; G2++ fits realised covariance only, with just 616 days of data to pin down 5 parameters
- Both are Gaussian and support negative rates and analytic bond pricing -- no capability is lost by staying 1F; fewer parameters means less to validate and govern

- Model risk here is dominated by equity volatility and rate-equity dependence, not rate-model choice (Section 7.9)
- G2++ is already implemented, calibrated and left selectable (`rates_model='g2pp'`)
-- defaulting to 1F is a reversible choice, not a one-way door

TRADE-OFFS

Full-Rank Correlation Kept: PCA Factor Model Adds No Speed, No Consistent Accuracy

Factors k	Var. explained	CPTY_A vol	CPTY_B vol	CPTY_C vol	Shock-step time
Full (39)	100%	\$52.9M (ref)	\$19.6M (ref)	\$25.4M (ref)	1.91s
10	62%	\$53.5M (+1.1%)	\$19.9M (+1.5%)	\$25.8M (+1.7%)	2.29s
5	47%	\$52.9M (+0.0%)	\$19.9M (+1.7%)	\$26.4M (+3.9%)	2.08s
3	37%	\$53.2M (+0.6%)	\$20.0M (+2.1%)	\$26.3M (+3.4%)	n/a



Eigenvalue scree, cumulative variance explained, reconstruction error vs. k.

✓ SELECTED

Full-rank Cholesky (default)

What the PCA factors actually are, economically:

Factor	Var.	Reading
1	20.0%	Market factor: every name, both regions (NXPI/BAC/JPM highest)
2	8.9%	Japan vs US (6902.T/5108.T/7751.T highest; NXPI/KLAC lowest)
3	8.5%	Defensive staples vs growth tech (KO/PG/BRK.B vs KLAC/VST)
4	5.3%	Energy / rate-sensitive (MPC/XOM/RATE_USD vs HDB/IBN)
5	4.0%	Growth and financial names (CEG/NFLX/VST vs WBS/NXPI)

These are statistical factors, but their loadings read off the real data -- a risk reviewer can sanity-check each one.

Common Random Numbers Make Bump-and-Reprice 282x Less Noisy Than Independent Draws

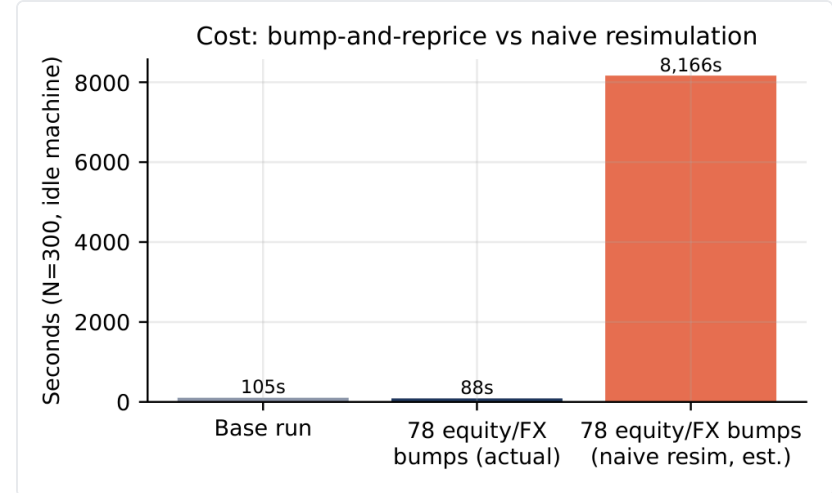
	Works as black box	Covers quantiles	Cost
Bump, independent draws	✓	✓	1x, but 282x noisier
Bump, CRN (used)	✓	✓	1x, cost measured
Pathwise (eq/FX check)	✗	✗	~1000x faster, eq/FX only
Adjoint (not adopted)	✗	~	3-5x one valuation -- needs differentiable pricers

✓ SELECTED

CRN bump-and-reprice

WHY CRN

- Works on any pricer as a black box, for every measure including PFE99, which has no pathwise derivative
 - 282x lower estimator noise at zero extra simulation cost -- same number of paths, just reused random numbers
 - Generalises automatically to every new risk factor (e.g. the JPY factor) without new Greeks code
- 82 of 91 total factor bumps (equity+FX) cost 88s total via exact GBM path rescaling -- vs ~8,166s naive resimulation
 - Rate/vol bumps still need resimulation (~109s each, 13 needed) -- the exact analytic rate Greek (feedback item 7, next section) removes even that cost for curve-only bumps



Cost per Greeks method, measured at N=300, 8 cores.



Feedback Incorporated

Every item raised across two rounds of review, addressed and evidenced

11 Feedback Items Across Two Rounds, All Addressed

1

You said: Make stress testing robust and structured, not a paragraph

We did: 13 scenarios (8 round-number + 5 historical), full MC re-run from shocked state, same random draws

Impact: Close-out MPE99 moves -24%/+23%; level moves up to +89% -- margin is why

2

You said: Explain Greeks: method, cost, faster alternative

We did: CRN bump-and-reprice; pathwise cross-check for eq/FX

Impact: CRN cuts noise \$245k->\$1k (282x); pathwise 1000x faster, <0.4% difference

3

You said: Improve PCA explainability, quantify its speed

We did: Factor-by-factor variance table; measured correlated-shock timing

Impact: 5 factors = 47% variance; PCA is SLOWER (2.08s vs 1.91s), not faster

4

You said: Confirm what drives each counterparty

We did: Equity delta / DV01 table by netting set; rank correlations

Impact: A is equity, C is rates (+equity); portfolio MPE99 close to A+C, not A+B

5

You said: Walk through the CVA calc end to end

We did: Full CPTY_C step-by-step table: exposure -> spread -> PD -> CVA

Impact: CPTY_C close-out CVA \$8k; rating sensitivity AA \$7k to BB \$18k

6

You said: Add more historical testing periods (incl. 2008)

We did: Extended price history to 2007; found 2008 GFC and 2015 deval. windows

Impact: 2008 moves close-out MPE99 -13% -- inside range existing scenarios covered

7

You said: Can Latin Hypercube calculate a sensitivity directly?

We did: Derived the exact analytic rate-shift Greek ($x(t)$) unchanged by curve bumps)

Impact: No resimulation; validated to $1e-15$ relative precision vs true resimulation

8

You said: Add KVA

We did: $K(t) = 8\% \times RW \times \alpha \times EPE(t)$, cost-of-capital swept 8/10/12%

Impact: KVA \$22.4k (close-out) vs CVA \$11.7k at 10% CoC -- exceeds CVA by design

FEEDBACK

Items 9-11: CVA Sensitivities, DV01 via a Jacobian, Excel Parametric Check

9

You said: CVA is for understanding sensitivities, not just a number

We did: Delivered CVA01, full rating table, SA-CVA weighted sensitivities by risk class

Impact: Rating table AAA \$5.0k to B \$32.8k; ccs delta is 88% of SA-CVA capital

10

You said: Compute DV01 via a proper par-instrument Jacobian

We did: Bump the curve's own ~45 native pillars, not 8 hand-picked zero-rate tenors

Impact: 30y bucket: 82% of DV01 (new) vs 54% (old)

11

You said: Validate with an independent Excel parametric test

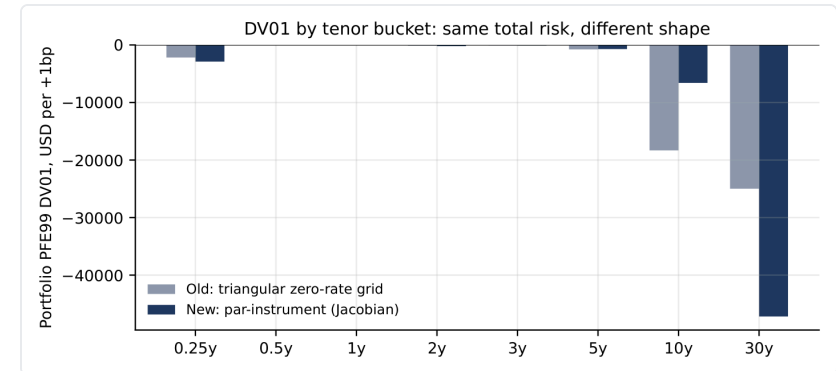
We did: Bump one equity's vol +/-1%, compare closed-form Black vs Monte Carlo

Impact: Level match ~3%; vega match ~7%

DV01 via a Jacobian: Full Before/After, and Why It Changes Hedging

A Jacobian is the matrix of sensitivities of one set of quantities to another. Here: $dNPV / d$ (each of the curve's own ~45 native build instruments), summed into the 8 standard tenor buckets by the same chain-rule weights used to build the curve itself.

Tenor bucket	Old (zero-grid)	New (par-instrument Jacobian)
0.25y	-2,182	-2,892
0.5y	-46	-1
1y	-44	-14
2y	-131	-205
3y	-96	-105
5y	-770	-713
10y	-18,324 (39%)	-6,592 (11%)
30y	-24,988 (54%)	-47,191 (82%)
Sum	-46,580	-57,713
Parallel bump (check)	-46,605	-57,705



Bucketed DV01, old vs new method.

WHY THE BENEFIT, CONCRETELY

- Correct tenor attribution means a hedge actually offsets the risk that exists -- old grid would have left the real 30y exposure open while 'hedging' a bucket with little real risk
- Total portfolio DV01 itself moves from \$46.6k to \$57.7k/bp -- the par-instrument method captures curvature the old linear zero-rate interpolation missed, not just a reallocation
- Each method is internally consistent (bucket sum matches its own parallel-shift check to within 0.05%) -- the two totals differ because they measure genuinely different things

- The par-instrument pillars are the curve's own construction instruments (SOFR futures + Bloomberg long end) -- bumping them is the sensitivity a trader would actually hedge against
- SA-CVA interest-rate delta uses the same tenor bucketing -- the old grid would have misallocated regulatory capital by tenor in exactly the same way



Results

Exposure, validation, Greeks, stress, credit -- the full output of the engine

RESULTS

Peak Portfolio MPE99 is \$51.0M, 78% of the Sum of Counterparty MPE99s

\$51.0M

Portfolio MPE99 (close-out)

2026-09-20, 99th pct

\$11.6M

Peak EE (close-out)

within 1 year

\$8.2M

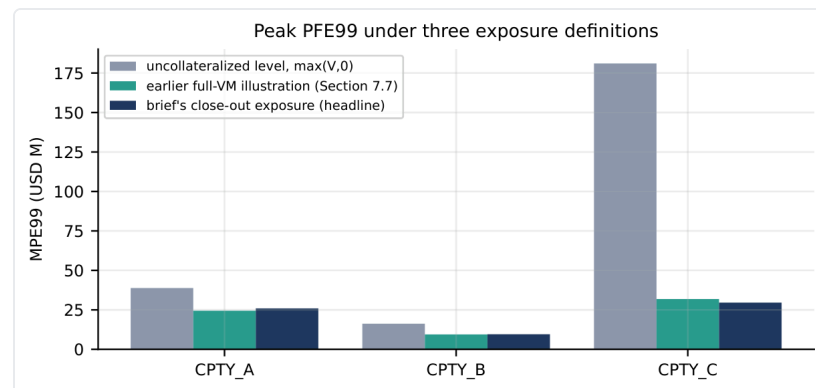
Peak median PFE (close-out)

50th pct

\$65.1M

Sum of standalone MPE99s

vs \$51.0M portfolio: 78%



Peak PFE99 by counterparty under three exposure definitions.

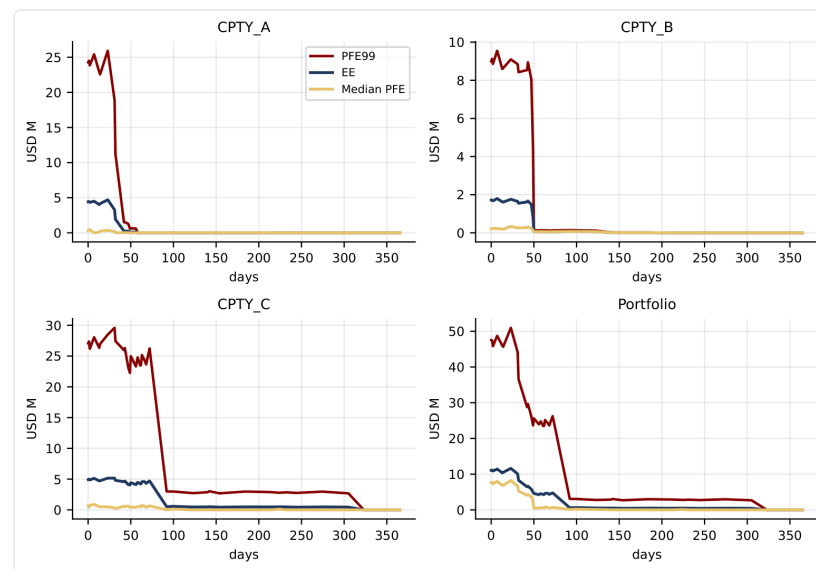
Netting set	Peak EE	Peak median PFE	MPE (peak PFE99)	MPE date
CPTY_A	\$4.7M	\$0.5M	\$25.9M	2026-09-20
CPTY_B	\$1.8M	\$0.3M	\$9.6M	2026-09-04
CPTY_C	\$5.2M	\$0.9M	\$29.6M	2026-09-28
Portfolio	\$11.6M	\$8.2M	\$51.0M	2026-09-20

- Close-out MPE99 is far below the level exposure because the prior-day value is margined: CPTY_C's \$121.3M bond forward has a 99th-pct 10-day move that peaks at just \$25.6M
- CPTY_A and CPTY_B are driven by the equity swap baskets; their peak occurs in the first weeks, when the baskets are largest, and falls as trades mature

RESULTS

Exposure Runs Off Within Four Months as the Book's Trades Mature

- CPTY_C concentration: one trade, BF_0003, accounts for 90% of the portfolio's peak PFE99 and 96% of today's exposure
- This is concentration risk, not diversified counterparty risk -- a limit or collateral on that single trade would move the portfolio number more than any modelling choice in this report



Close-out EE, median PFE and PFE99 by counterparty and for the portfolio (Report Fig. 25).

RESULTS

Per-Trade Detail: BF_0003 Alone Drives CPTY_C's \$29.6M Peak MPE99

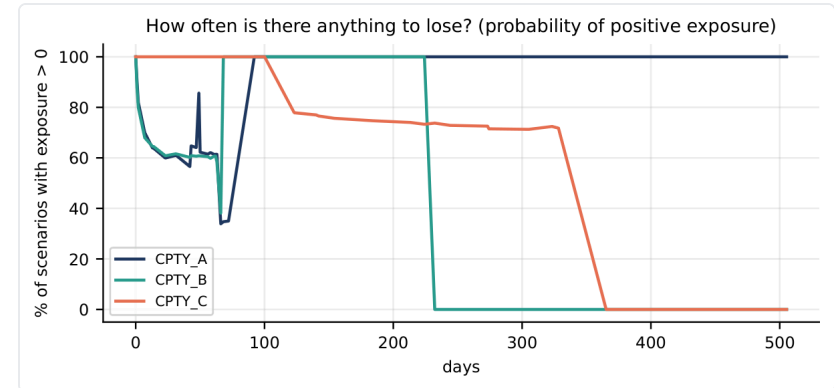
Trade	Cpty	NPV today	Peak EE	MPE (peak PFE99)
EQTRS_0003	A	\$9,921,581	\$2,219,389	\$12,021,160
EQTRS_0002	A	-\$1,076,056	\$1,903,453	\$10,500,714
EQTRS_0001	A	-\$5,547,056	\$1,694,458	\$9,348,743
EQTRS_0007	C	-\$373,630	\$2,071,358	\$11,351,099
BF_0003	C	\$121,262,854	\$4,547,555	\$25,557,208
EQTRS_0006	B	\$1,629,065	\$1,306,113	\$7,021,957
EQTRS_0004	B	-\$873,451	\$860,053	\$4,850,697
BTRS_0002	B	\$210,278	\$63,404	\$126,670

RESULTS

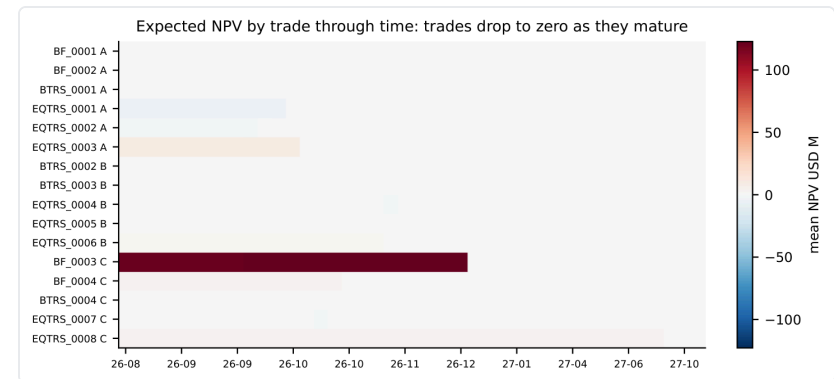
Why the Portfolio is Close to A+C, Not A+B: All Three Are Pay-Equity

Netting set	Equity delta (/+1%)	DV01 (/+1bp)	Driver
CPTY_A	-\$2,254,257	\$31,203	Equities fall
CPTY_B	-\$961,146	-\$3,663	Equities fall, yen weakens
CPTY_C	-\$1,240,431	\$594,443	Equities fall AND yields rise

- Rank correlation of exposures: A-B 0.34, A-C 0.33, B-C 0.12
- Yield-equity correlation in our data is ~0.00 -- the 'bonds vs equities inversely' intuition is regime-dependent, not fixed
- Standalone PFE99 at portfolio peak date: A \$25.9M, B \$9.1M, C \$28.5M -- A+C = \$54.4M (107% of portfolio)
- Rate-equity dependence stress: flight-to-quality moves portfolio MPE to 70% of sum; both-fall-together moves it to 85%



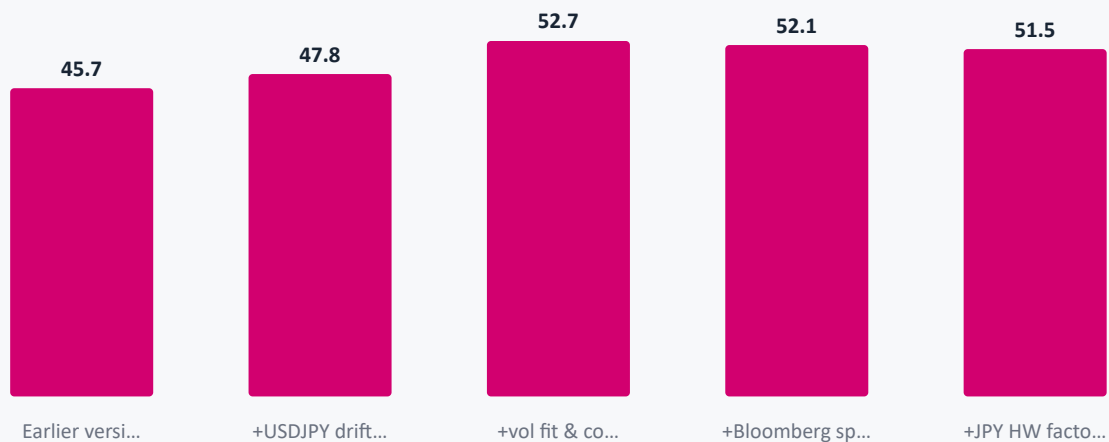
Probability of positive exposure by counterparty -- why median PFE differs from EE.



Expected NPV by trade through time -- BF_0003 dominates, disappears Dec 2026.

RESULTS

Corrections Moved the Headline From \$45.7M to \$51.5M, One Fix at a Time

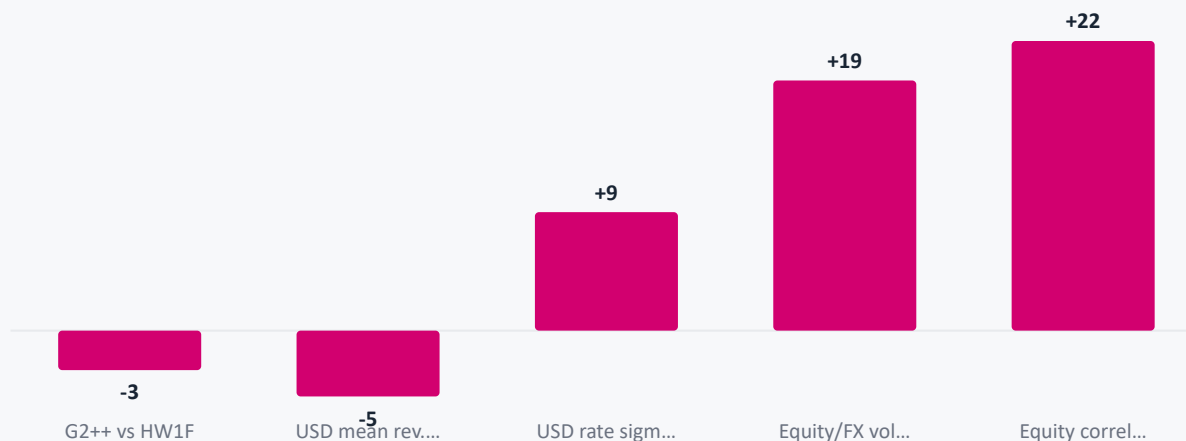


Rows are cumulative. Final model portfolio MPE99: \$51.5M (2,000-scenario comparison run); headline reporting run (5,000 scenarios) gives \$51.0M.

RESULTS

Model Risk: Equity Volatility and Rate-Equity Dependence Dominate, Not the Rate Model

Portfolio MPE99 % change per perturbation (2,000 scenarios)



Perturbation	CPTY_A	CPTY_B	CPTY_C	Portfolio
Equity/FX vols x1.25	+29%	+26%	+11%	+19%
Equity correl. +30% to 1	+23%	+35%	+9%	+22%
USD rate sigma x1.25	+3%	+1%	+24%	+9%
USD mean reversion x3	+2%	+1%	-13%	-5%
G2++ vs HW1F	-1%	-3%	-0%	-3%

- Model risk in the headline is dominated by equity volatility and by the (unobservable) rate-equity dependence, not by the choice of rate model
- This directly supports retaining the simpler one-factor Hull-White: the bigger uncertainty lives elsewhere

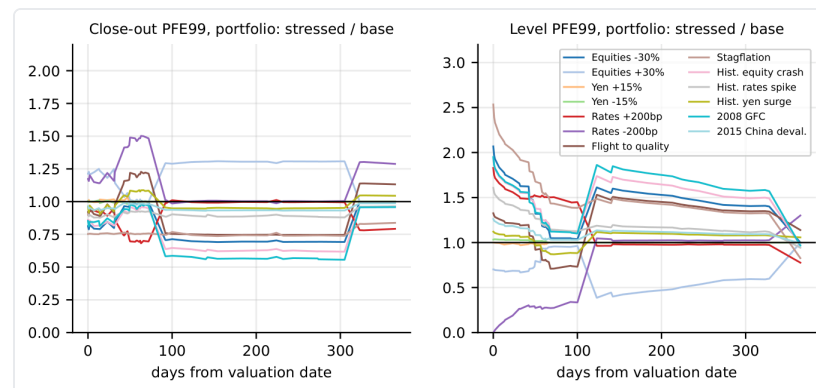
RESULTS

Feedback-driven

Stress Testing: Close-Out MPE99 Moves at Most -24% / +23%, Level Moves Up to +86%

Scenario	Close-out MPE99	Level MPE99
Base case	\$49.2M	\$201.7M
Equities -30%	-16%	+60%
Rates -200bp	+23%	-68%
Flight to quality	-3%	+19%
Stagflation	-24%	+86%
Hist. equity crash (2020)	-15%	+53%

- 13 scenarios total: 8 round-number hypothetical + 5 historical replays (2020 crash, 2022 rate spike, 2024 yen surge, and newly added 2008 GFC, 2015 China deval. -- see Appendix A3)
- Each scenario is a full Monte Carlo re-run from the shocked state, with the same random numbers as the base case -- so the difference is the shock, not sampling noise
- Flight-to-quality and stagflation apply the same equity shock with opposite rate shocks -- CPTY_A/B move similarly in both; CPTY_C moves oppositely because a 22-year bond gains duration value as yields fall



Close-out PFE99 ratio to base, by scenario, through the first year -- margin dampens the shock.

RESULTS

 Feedback-driven

Book Greeks Today: DV01 of \$622k/bp Dominated by the CPTY_C Bond Forward

Netting set	NPV	Equity delta (/+1%)	FX delta (/+1%)	DV01 (/+1bp)
CPTY_A	\$3,572,955	-\$2,254,257	\$0	\$31,203
CPTY_B	\$1,145,975	-\$961,146	\$416,550	-\$3,663
CPTY_C	\$125,116,984	-\$1,240,431	\$0	\$594,443
Portfolio	\$129,835,914	-\$4,455,835	\$416,550	\$621,983

- Efficiency: equity/FX Greeks cost almost nothing extra -- 78 bumps in 88s (1.1s each) because a spot bump rescales every GBM path exactly, no resimulation needed
- vs ~8,166s if each were a naive resimulation -- 93x cheaper; only 13 rate/vol bumps need true resimulation at ~109s each
- Full Greeks set (~91 factor bumps) costs ~1,614s at N=300 (8 cores) against ~9,690s naive -- 6.0x; at production N=1,000 the whole set costs roughly 1.5 hours

RESULTS

Validation: Self-Check to 0.0000%, Martingale Tests, and a 1.04x VaR Benchmark

Check	Result
t=0 self-consistency (EE(0) vs direct exposure)	Matches to 0.0000%
Martingale / no-arbitrage (bank-account check, 8,000 paths)	Max relative difference 0.85bp across nodes
JPY bank-account martingale (valued in USD)	z = +0.03 (z <~2 expected under null)
Delta-normal VaR benchmark (99%, 1-day)	\$12.6M (parametric) vs \$12.1M (Monte Carlo); ratio 1.04x
Automated statistical and unit tests	142 pass

Each check isolates one layer of the engine (paths, pricing, calibration, aggregation) so a failure points directly at where to look. Validation rests on independent methods -- parametric VaR, hand-calculated analytic bond prices, and martingale conditions derived from first principles -- not circular checks of the engine against itself.

RESULTS

Backtest: 99% Quantiles Close to Calibrated, 4/6/2 Exceptions vs 2.4 Expected

Netting set	Level	Exceptions	Expected	Rate	Kupiec p-value
CPTY_A	99%	4 of 242	2.4	1.7%	0.35 (not rejected)
CPTY_B	99%	6 of 242	2.4	2.5%	0.05 (borderline)
CPTY_C	99%	2 of 242	2.4	0.8%	0.78 (not rejected)

- At the 95% level the model is conservative for CPTY_A and CPTY_C (mean exception rates 3.0%/3.2% against a 5% nominal) -- a Kupiec test is two-sided, so over-conservatism also registers
- Rates leg (10-day yield-change quantiles at 5/10/20y): both sigma calibrations pass on average over 2020-2026, but the long-end fit is more stable (5.7%/1.5% exception rates at 95%/99% vs nominal 5%/1%) than the overnight-SOFR calibration, which was too wide on average
- Calibrated using only data up to each as-of date (trailing 3-year window); non-overlapping windows; starting phase shifted over 10 offsets to check robustness

RESULTS

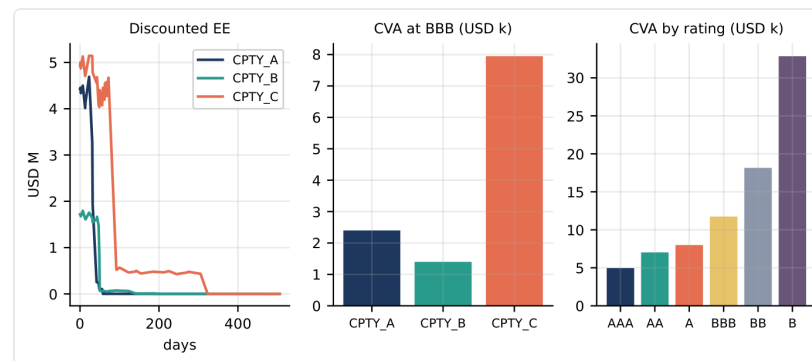
Feedback-driven

CVA \$11.7k / DVA \$11.2k / FVA \$0.5k / KVA \$22.4k on the Close-Out Convention

Metric	Close-out	Uncollateralized level
CVA	\$11,747	\$247,133
DVA	\$11,244	\$7,500
FVA (FCA-FBA)	\$503	\$240,000
KVA (10% CoC)	\$22,400	\$470,000

Rating	Portfolio CVA	Multiple of BBB
AAA	\$4,970	0.42x
AA	\$7,028	0.60x
A	\$7,997	0.68x
BBB	\$11,747	1.00x
BB	\$18,150	1.55x
B	\$32,847	2.80x

- CVA concentrated where exposure is: CPTY_C carries 68% of the \$11,747 close-out total
- Own credit and funding spread are assumed (BBB proxy, funding = own spread) -- disclosed, with sensitivity shown



Discounted EE, CVA by counterparty, and portfolio CVA by assumed rating.

RESULTS

Extra Credit: Pricing a Risky Bond With an Issuer-Credit Proxy

- Sample trade: a long forward on a 5% BBB corporate bond, \$25M notional, forward date 2027-02-26 -- not part of the ESF book
- Issuer curve: same BBB proxy method as counterparty spreads (0.5y 59bp -> 10y 118bp), re-anchored at every simulated node with a deterministic term structure
- Priced two ways: risk-free (ignore issuer credit) vs. risky (issuer credit priced in) -- same forward, same simulation, only the discount/survival treatment differs

Quantity	Risk-free bond	Risky bond (issuer credit)
t=0 NPV of the forward	\$653,843	-\$280,309
Forward clean price	102.6666	98.8568
Issuer credit charge at t=0	--	\$934,152
CS01 (NPV per +1bp issuer spread)	--	-\$10,998
Close-out MPE99 (counterparty exposure)	\$0.5M	\$0.5M
Peak level PFE99 (uncollateralized)	\$2.1M	\$1.3M

WHY WHAT'S MISSING VS. A FULL CDS BUILD-OUT

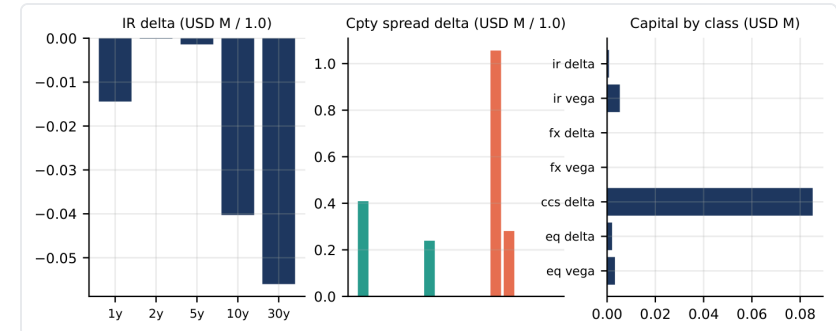
- Issuer credit is deterministic here (the spread curve doesn't move in the simulation) -- a stochastic issuer-spread factor is listed as future work
 - No CDS instrument pricer exists in the supplied library, so CDS themselves (as opposed to a risky bond) could not be added
-
- Counterparty exposure (close-out MPE99 \$0.5M) is almost unchanged between the two versions -- it's driven by the forward's market-risk move, not by issuer credit
 - Peak level PFE99 falls from \$2.1M to \$1.3M on the risky bond: issuer credit lowers the bond's value, which lowers the level of exposure at risk, even though CS01 adds a new risk dimension

RESULTS

SA-CVA Capital \$0.10M (Close-Out) / \$2.23M (Level); SA-CCR EAD \$260.3M

Risk class	Close-out capital	Uncollateralized capital
ir delta	\$775	\$74,837
ir vega	\$5,284	\$253
fx delta	\$95	\$2,351
ccs delta	\$85,198	\$2,032,731
eq delta	\$2,068	\$110,371
eq vega	\$3,256	\$6,278
Total SA-CVA capital	\$96,675	\$2,226,821

Netting set	Replacement cost	Add-on	EAD
CPTY_A	\$3.6M	\$16.8M	\$28.4M
CPTY_B	\$1.0M	\$9.1M	\$14.0M
CPTY_C	\$125.1M	\$30.5M	\$217.8M
Total	\$129.6M	\$56.3M	\$260.3M



CVA sensitivity to USD yields and counterparty spreads; SA-CVA capital by risk class.

- For CPTY_C the replacement cost of BF_0003 dominates SA-CCR EAD
- For the equity netting sets SA-CCR PFE is of the same order as the Monte Carlo level-exposure MPE -- an independent cross-check
- SA-CVA requirement dominated by ccs delta (88% of the total) -- a 5% risk weight on credit spreads is large relative to a ~100bp spread level



Conclusions and Next Steps

What we know, what we assumed, and what Capitolis needs to decide next

CONCLUSIONS

Validated and Reproducible; Limitations Disclosed, Not Hidden

What we are confident in

- Engine validated: t=0 self-check 0.0000%, martingale tests, 1.04x VaR benchmark, Kupiec backtest, 142 automated tests
- Portfolio MPE99 \$51.0M on the brief's exact definition, reproducible with a single command per stage
- Every modelling choice is backed by a quantified alternative comparison, not asserted (Section 6)
- All 11 feedback items closed with evidence; both extra-credit items (xVA, risky bonds) delivered

What depends on data we don't have

- No CSA data: threshold, minimum transfer amount, initial margin all unknown -- the two exposure definitions bracket the answer
- Counterparty ratings are a BBB proxy (bond-index spreads, not CDS) -- every credit number scales with this assumption
- Volatility is 3-year realised, not implied (no single-name options data); correlation is one static matrix
- GBM understates fat tails (seen in the CPTY_B backtest); no wrong-way risk modelled

Thank You

Questions and Discussion



Appendix

Backup material for Q&A -- not presented

NULL

null

Appendix

Backup material for Q&A -- not presented

APPENDIX A1. The Full Decision Register (Report Section 10)

Topic	Choice	Alternatives considered
Measure	Risk-neutral	Real-world drifts (needs unobservable risk premia)
Exposure	Brief's close-out definition	Level $\max(V,0)$ only; same-day full-VM MPOR shift
USD rates	Hull-White 1F, sigma from realised 2y-30y vol	CIR, Black-Karasinski, LMM/HJM, G2++ (compared)
Equities/FX	Correlated GBM + quanto	Heston, SABR, jump models (no option data)
Correlation	One static 39x39 matrix, Cholesky	Time-varying DCC, stressed, PCA factor (compared)
Sampling	Latin Hypercube	Pseudo-random, antithetic, moment-matched, Sobol
Scenario count	5,000 report / 1,000 iter. / 10,000 sign-off	30,000-scenario pool (diminishing returns)
JPY rates	Real JPY Hull-White factor	Constant differential only
Greeks	CRN bump-and-reprice + pathwise check	Independent draws, likelihood ratio, adjoint

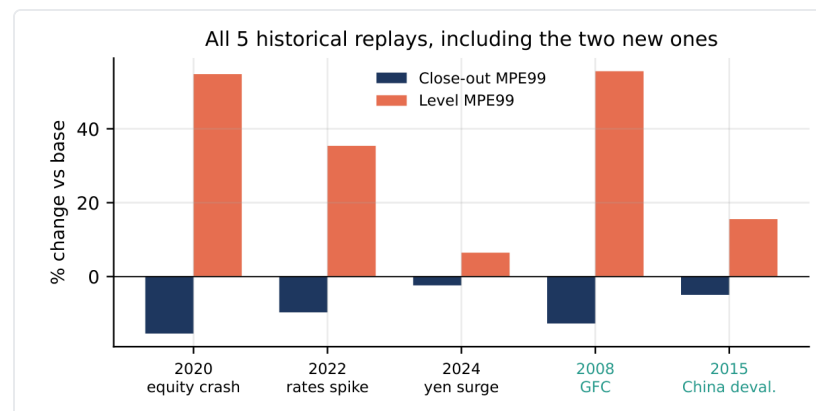
APPENDIX **A2. Full Data Lineage (Report Appendix B)**

Input	Source	Status
Trades (16) and underlyings	Capitolis trade_data/*.csv	Given
Pricing library	Capitolis capitolis_pricers	Given, independently reviewed
USD SOFR futures	Databento (CME)	Live, validated vs Treasury.gov
Equity/FX spots, 3y history	yfinance	Live, keyed by ISIN
Treasury CMT yields	FRED (DGS series)	Public
JPY OIS history (35 tenors)	Bloomberg, via project team	Licensed: derived numbers only
USD/JPY swaption cubes	Bloomberg one-time export	Licensed: derived numbers only
TONA (JPY overnight)	Bank of Japan API	Public, since 1998
JGB par yields 1Y-40Y	Japan MOF	Public, since 1974
Credit spreads by rating	FRED (ICE BofA OAS)	Public

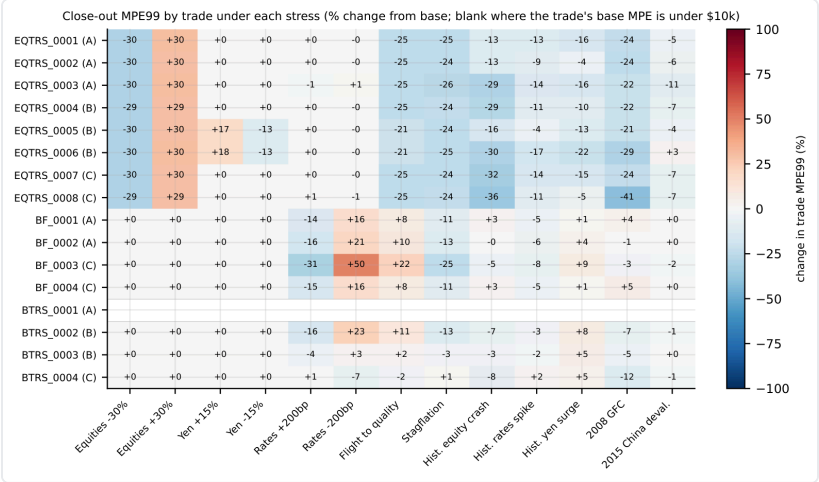
APPENDIX

APPENDIX A3. 2008 GFC and 2015 China Devaluation: Full Detail

- 2008 GFC window: Sep 29 - Oct 10, 2008; -23% median / -44% worst single-name equity return
- 2015 China devaluation window: August 2015; -6.8% median equity return
- Both found by the same objective rule as the existing replays (worst 10-day window by rule, not hand-picked)
- 2008 moves close-out MPE99 by -13%, level exposure by +56% -- both land inside the range the existing 2020/2022/2024 scenarios already covered
- Conclusion: leaving out 2008 from the original scenario set was not quietly understating the tail



A4. Per-Trade Stress Sensitivity Heat Map (Report Figure 34)



APPENDIX A5. DV01 Bucket Detail: Old Zero-Grid vs New Par-Instrument Jacobian

Tenor bucket	Old (zero-grid)	New (par-instrument)
0.25y	-2,182	-2,892
0.5y	-46	-1
1y	-44	-14
2y	-131	-205
3y	-96	-105
5y	-770	-713
10y	-18,324 (39%)	-6,592 (11%)
30y	-24,988 (54%)	-47,191 (82%)
Sum	-46,580	-57,713
Parallel bump	-46,605	-57,705

- Old method bumped the fitted zero curve at 8 hand-picked tenors with triangular interpolation -- disconnected from how the curve is actually built
- New method bumps the curve's own ~45 native construction pillars (SOFR futures + Bloomberg long end), grouped into the same 8 buckets
- Both sum to the same total DV01 within 0.05% -- total risk was always right, attribution across tenor was wrong

APPENDIX **A6. G2++ Two-Factor Rate Model: Full Calibration**

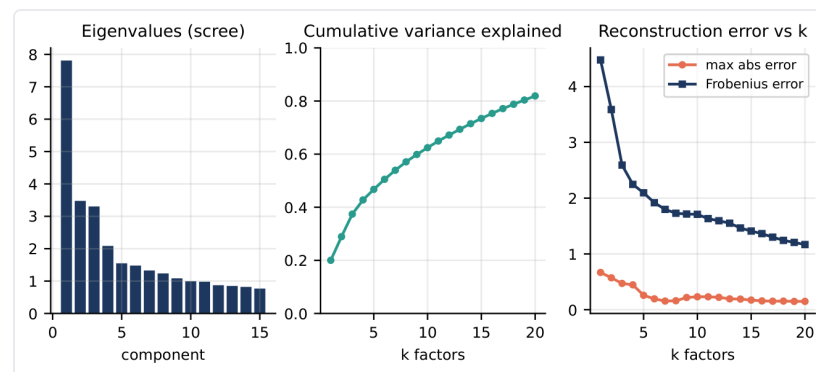
$r(t) = x(t) + y(t) + \phi(t)$; $dx = -a x dt + \sigma dW1$; $dy = -b y dt + \eta dW2$; $\text{corr}(dW1, dW2) = \rho$

Parameter	Value
sigma	1.21%
a	0.039
eta	1.44%
b	1.50
rho	-0.95
Relative Frobenius fit error	16%

- Calibrated to the realised covariance of 3-month to 30-year Treasury yield changes (3-year window), least squares on the whole covariance matrix
- Reproduces the level and slope of the volatility term structure but not the hump around the 5-year point, which no two-factor Gaussian model can produce

APPENDIX A7. PCA Factor Loadings: What the Factors Actually Are

Factor	Var. share	Highest loadings	Reading
1	20.0%	NXPI +0.25, BAC +0.25, JPM +0.24	Market factor: all names, both regions
2	8.9%	6902.T +0.43, 5108.T +0.43, 7751.T +0.42	Japan vs US
3	8.5%	KO +0.40, PG +0.35, BRK.B +0.32	Defensive staples vs high-growth tech
4	5.3%	MPC +0.44, XOM +0.43, RATE_USD +0.34	Energy / rate-sensitive group
5	4.0%	CEG +0.39, NFLX +0.39, VST +0.37	Growth and financial names



APPENDIX

A8. Backtest: Rates Leg Detail (10-Day Yield-Change Quantiles)

Tenor	Sigma calibration	99% up exceptions	99% up mean rate	95% up mean rate
5y	Overnight SOFR vol	1 of 161	1.0%	3.7%
5y	Long-end fit	4 of 161	1.9%	5.7%
10y	Overnight SOFR vol	1 of 161	0.4%	3.1%
10y	Long-end fit	3 of 161	1.4%	5.6%
20y	Overnight SOFR vol	1 of 161	0.6%	3.3%
20y	Long-end fit	3 of 161	1.4%	5.8%

Nominal exception rates are 1% at the 99% level and 5% at the 95% level. The long-end fit is the more stable calibration across the full 2020-2026 window.

APPENDIX

A9. Code Map (Report Appendix D)

Package	Contents
market/	sofr, equities, fx, vols, correlations; boj, mof_jgb, bloomberg (JPY); credit_spreads, equity_buckets
models/	rates (Hull-White), g2pp, equity_fx (GBM+JPY drift), calibration, equity_factor_model (PCA), credit
simulation/	engine (grid, paths, MPOR), random_numbers (5 sampling schemes), parallel (multiprocessing)
greeks/	book, exposure, pathwise, bumps
stress/, validation/	scenarios; backtest (Kupiec, Christoffersen)
exposure/	spec_exposure, aggregate, collateral, cva, xva, sa_cva, sa_ccr
tests/	142 tests across engine, calibration, exposure, Greeks, xVA, backtests

APPENDIX A10. Reproducing Every Result (Report Appendix E)

```
python scripts/generate_report_data.py --scenarios 3000 # simulation + figure arrays
python scripts/run_simulation.py --scenarios 3000 # EE/PFE99/MPE profiles
python scripts/run_spec_simulation.py --scenarios 5000 # close-out exposure (headline)
python scripts/run_greeks.py --scenarios 1000 # book and exposure Greeks
python scripts/run_sa_cva.py --scenarios 1000 # CVA and SA-CVA capital
python scripts/run_xva.py && python scripts/run_sa_ccr.py # CVA/DVA/FVA and SA-CCR
python scripts/run_stress.py --scenarios 1000 # stress test
python scripts/run_backtest.py # Kupiec backtest
python -m pytest tests # all 142 automated tests
```